INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION

TRUCK, CLEANER, CATCH BASIN, VACUUM-PRESSURE JET, DED

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers truck-mounted vacuum catch basin and sewer cleaners which use a high-velocity-pressure jet for removing sludge and debris from drainage lines, and simultaneously vacuums the sludge and debris from the catch basin or manhole.

1.2 Classification. The cleaners furnished shall be of the following types and sizes (see 6.2 and 6.3).

Type I - For 25 foot depth operation.

Size 1 - With a minimum 8 useable cubic yard (YD^3) body and a minimum length of 112 inches.
Size 2 - With a minimum of 14 useable YD^3 body and a minimum length of 144 inches.

Type II - For up to 100 foot depth operation.

Size 2 - With a minimum 14 useable YD^3 body and a minimum length of 144 inches.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 2320

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

W-B-131 - Battery, Storage: Vehicular, Ignition, Lighting and Starting.

STANDARDS

FEDERAL

FED-STD-297 - Rustproofing of Commercial (Nontactical) vehicles.
FED-STD-595 - Colors.

MILITARY

MIL-STD-281 - Automobiles, Trucks, Truck-Tractor, Trailer and Trailer Dollies, Preservation and Packaging of.

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS shall be the issue of the non-Government documents which is current on the date of the solicitation.

ENVIRONMENTAL PROTECTION AGENCY (EPA)

Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines: Certification and Test Procedures.
Interstate Motor Carrier Noise Emission Standards.
Motor Vehicle Air Pollution Standards.

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(Application for copies should be addressed to the Public Affairs Office, Environmental Protection Agency, Rockville, MD 20852; or CFR, Title 40 should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

DEPARTMENT OF LABOR (DoL)
OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

Title 29, code of Federal Regulations, Chapter XVII, Part 1910, and Amendments.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

DEPARTMENT OF TRANSPORTATION (DoT)

Federal Motor Vehicle Safety Standards and Regulations.
Federal Motor Carrier Safety Regulations.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A167 - Stainless and Heat Resisting Chromium-Nickel Steel, Steel Plate Sheet and Strip Specification for.
A224 - High-Strength, Low-Alloy, Structural Steel.
A588 - High-Strength, Low-Alloy, Structural Steel with 50,000 lb/in^2 Minimum Yield Point to 4 Inches Thick..

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

STATE OF CALIFORNIA

California Vehicle Code.

(Application for copies should be addressed to the Department of Motor Vehicles, 2570 24th Street, Sacramento, CA 95809.)

TIRE AND RIM ASSOCIATION, INC. (TRA)

TRA Yearbook.

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(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The truck catch basin cleaner, referenced to hereinafter as "cleaner" shall be a commercial product of the manufacturer's latest design and shall meet the requirements specified herein. The cleaner shall consist essentially of a diesel-engine-driven, commercial truck which has a mounted water tank, a body containing dust control equipment, a rear door, and dump hoist or dump mechanism. The cleaner shall be equipped with hydraulic boom, hose reel and hose vacuum system, high pressure pump, and all necessary equipment to make a complete and operable cleaner. All equipment shall be driven by diesel power and shall be operable independently or simultaneously. Provisions shall be made on the cleaner for storage of all equipment and tools. The cleaner shall be capable of continuous warranted operation when vacuuming either wet or dry material.

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.2.1 and 6.6).

3.3 Standard commercial product. The cleaner shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the cleaner being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible

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without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.5 Safety. All rotating or reciprocating parts that are of such nature or so located as to be a hazard to operating of maintenance personnel shall be enclosed or properly guarded in accordance with Code of Federal Regulations, Chapter XVII, Part 1910, paragraph 1910.219 in effect at time of manufacturer.

3.6 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.7 DoT motor vehicle safety standards. The vehicle shall comply with all applicable Motor Vehicle Safety Standards in effect on date of manufacture.

3.8 Air pollution control. Trucks shall comply with the EPA Regulations governing Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines in effect on the date of manufacture. In addition, trucks destined for California shall comply with State of California regulations governing air pollution control in effect on the date of manufacturer.

3.9 Noise. If the noise level in the area occupied by the operator(s) exceeds 90 decibles (db(A)), when operating the cleaner in the combined pressure-jetting and/or vacuum mode, a caution plate shall be permanently posted on the cleaner in a conspicuous protected location and shall be clearly visible and legible to all personnel exposed to excessive noise levels. The caution plate shall read: CAUTION - HEARING PROTECTION REQUIRED. The plate shall have a yellow background with black lettering and plate shall be made of corrosion-resistant material.

3.10 Performance.

3.10.1 Pump. A minimum of 45 gallons of water per minute (gal/min) at 2,000 pounds per square inch gage (lb/in^2g) shall be provided at the pump discharge for removing sludge and debris from drainage lines.

3.10.2 Operation. Type I and type II cleaners shall clean an 8-inch unobstructed sewer line that is 25-feet deep and 400-feet long in not more than 17 minutes. During the hydraulic-rodding with the high pressure hose, the hose shall be self-propelling for the full 400-feet and not require manual assistance.

3.10.3 Blower-compressor. With a vacuum nozzle not larger than 8 inches in diameter, the blower-compressor shall be capable of depositing not less

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than $2YD^3$ of sand into the body under the conditions in table I. At any temperature range between -10 degrees (°) to 90° Fahrenheit (F). Retained volume of dry sand shall not be less than 95 percent weight.

TABLE I. Material deposited in body.

A. TYPE I - REACH

Depth below ground level	Horizontal distance	Material condition	Maximum time allowed per cubic yard
25 feet	12 feet	Wet 1/	6 minutes
25 feet	12 feet	Dry 1/	4 minutes

B. TYPE II - REACH

Depth below ground level	Horizontal run	Material condition	Maximum time allowed per cubic yard
100 feet	40 feet	Wet 1/	5 minutes
100 feet	100 feet	Dry 1/	7 minutes

1/ Average density pounds per cubic foot (lb/ft^3).

Dry sand 90 - 150 lb/ft^3.
Wet sand 60 - 126 lb/ft^3.

3.10.4 Loading light material. On size 2 only, load light dry material such as leaves, saw dust, etc., to the manufacturer's advertised body capacity, but not less than the minimum usable capacity specified in 1.2 with dust emissions not to exceed requirements of 3.12.1.

3.10.5 Truck. The cleaner in the loaded with full payload and with all equipment, shall ascend a minimum grade of 2.0 percent at not less than 50 miles per hour with the transmission in direct drive.

3.11 Truck construction.

3.11.1 Chassis. Chassis shall be 4 x 2 for size 1 and 6 x 4 for size 2.

3.11.2 Gross vehicle weight rating. The gross vehicle weight rating (gvw) shall be not less than the following:

Size 1 = 37,000 pounds (lb).
Size 2 = 55,000 lb.

3.11.3 Engine. The truck shall be equipped with a liquid cooled, compression ignition, two or four stroke cycle diesel engine, with not less than six cylinders, having horsepower (hp), torque, and speed characteristics to

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satisfactorily perform the power and travel requirements. When chassis engine is utilized for both travel and auxiliary power functions, it shall have hp rating sufficient to perform both vacuuming and water pumping simultaneously without exceeding 90 percent of governed RPM. If an ether starting system is furnished in lieu of a glow plug, it shall be of the measured shot type, controlled from the driver's compartment, with a reservoir of not less than 12 fluid ounces, and shall be inoperative with the engine warm.

3.11.3.1 Oil filter. A full flow oil filter shall be furnished.

3.11.3.2 Governor. Engine governor shall be furnished and set and sealed to limit engine to manufacturer's maximum recommended operating speed.

3.11.4 Fuel system. Fuel system shall conform to DoT Motor Carrier Safety Regulations, Section 393.65. Fuel tank shall have sufficient capacity for 8 hours of normal operation (2 hrs commuting and 6 hrs full loading pumping).

3.11.5 Air cleaner. A dry type engine intake air cleaner shall be furnished, and provided with a restrictor indicator gage mounted in the cab or visible from the control station.

3.11.6 Electrical system. The electric system shall be 12 volt (V). Alternator shall have a minimum rated capacity of 80 amperes (A). All lights, reflectors, and wiring shall conform to DoT Motor Carrier Safety Regulations.

3.11.7 Storage battery(s). The storage battery(s) normally furnished to make a complete electrical system shall be of the maintenance free type having the maintenance free characateristics listed in W-B-131.

3.11.8 Steering. Truck shall be equipped with full power hydraulic steering.

3.11.9 Transmission. Truck shall be equipped with an automatic transmission having not less than four forward and one reverse speeds. When the transmission is utilized in combination with P.T.O. drive(s) for powering the cleaner in components, there shall be remote controls furnished at the cleaner external control panel for engagement and disengagement.

3.11.10 Brakes. Brakes shall conform to Federal Motor Carrier Safety Regulation 393.40 through 393.43 and 393.45 through 393.52.

3.11.11 Suspension. Truck shall be equipped with suspension system having components with a rated capacity at least equal to the loads imposed on each member, measured at the ground, with the truck load to rated gvw. When suspension capacity is rated at the spring pads, unsprung weight shall be deducted. When available from chassis manufacturer as standard or optional equipment, shock absorbers of the heavy duty, double-acting, hydraulic type shall be mounted on the front axle.

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3.11.12 Axles. Front and rear axles shall be manufacturer's standard full floating type. Axle ratings shall be at least equal to the load imposed on each axle, measured at the ground, with truck loaded to rated gvw.

3.11.13 Wheels and tires. Chassis shall be equipped with single front and dual rear disk type wheels shall be provided. Wheel and tire rating shall conform to T&RA recommendations. Tires shall have rated capacity at least equal to the load imposed on each tire, measured at each wheel, at the ground, with truck loaded to rated gvw.

3.11.14 Windshield washers and wipers. The cab shall be equipped with dual windsheld washers and dual windshield wipers. Windshield wipers shall be multispeed type and operated by either air or electric motor(s).

3.11.15 Cab heater and defroster. Heater shall be manufacturer's standard liquid coolant operated type, meeting the requirement of DoT Motor Carrier Safety Regulations, section 393.77, as applicable. Defroster shall keep not less than 75 percent of windshield area clear at an ambient temperature of -25°F and shall be provided with an air regulating control. Heater and defroster controls shall be mounted within easy reach from the seated operator's position.

3.11.16 Revolving warning light. A revolving warning light shall be provided and mounted with reinforcement at point of attachment on vehicle centerline in a position affording 180° visibility of the full sealed beam lamps. The warning light shall be operated by means of a separate switch, with a red indicating light, located on instrument panel or dash and marked to indicate function of switch. The warning light shall have a clear dome, shall have not less than two sealed beam lamps each having a minimum of 35,000 candle power and a minimum flash rate exceeding 80 flashes per minute. The lamps shall be amber in color.

3.11.17 Accessories and equipment for trucks. Chassis equipment shall be complete with all accessories furnished as standard equipment by the manufacturer. The following minimum equipment shall be furnished:

a. Outside rearview mirrors, west coast type, mounted on each side of cab or body as applicable.
b. Key-operated ignition switch.
c. Ammeter, charging, or battery voltage indicator.
d. Fuel gage.
e. Oil pressure gage or indicator.
f. Speedometer with recording odometer.
g. Dual sunvisor.
h. Front door mounted armrest on driver's side.
i. Driver's compartment ventilator other than window.
j. Horn.

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3.12 Components for sewer and catch basin cleaning.

3.12.1 Body. The body shall have a minimum usable capacity in YD^3 as specified in 1.2, and be a maximum of 96 inches wide. The floor shall be not less than 3/16 inch steel (U.S. revised standard gage No. 7). The sides and roof of the rectangular bodies shall be as a minimum, constructed with not less than 0.1196-inch (U.S. revised standard gage No. 11) steel cylindrical bodies shall be constructed of not less than 3/16 inch steel. The inside of the body shall be smooth, crevice free, and manufactured from a corrosion-and abrasion-resistant copper bearing high strength steel suitable for welding equal to ASTM A242 or ASTM A588. When specified (see 6.2), the inside of the body shall be manufacturered from a corrosion-resistant steel compatible with salt water environment equal to ASTM A167, type 316. The body frame shall be designed and built for loads required for the body capacity. The body shall have a full rear opening door with replaceable neoprene type seal. A hydraulic dump hoist shall be provided to remove debris from the body or a hydraulically powered dump mechanism. If provided, the hydraulic dump hoist shall raise the body with rated load to at least 50° from horizontal and the body shall be provided with a drain of not less than 5 inches in diameter and consist of a flexible hose with storage rack. On size 1 unit, air baffles shall be provided to the limit carryover of material into the air stream. On size 2 unit, the cleaner shall be provided with means of separating and removing 95 percent of all dust particles larger than 50 microns in diameter from the discharge air stream, when loading any type of dry or wet material.

3.12.2 Water tank. An auxiliary tank having a capacity of not less than 1,000 gallons for size 1 unit and 1,500 gallons for size 2 unit shall be provided as an integral part of the cleaner. The tank shall be constructed of corrosion-and abrasion-resistant copper bearing high strength steel suitable for welding equal to ASTM A242 or ASTM A588 reinforced by steel channels, and equipped with antisurge baffles. In lieu of steel constructed tank, poly-ethlene tank(s) may be provided. The interior of the water tank shall be provided with a coating of black asphalt varnish or rust inhibiting material. The water tank shall be filled from a hydrant through a 2-1/2 inch hose and antisyphon device mounted on the curb side of the tank. The cleaner shall be provided with 15 feet of 2-1/2 inch fire hose with couplings and provisions for storage. The water tank shall not be connected to a waste water recycling system.

3.12.3 Separator. A separator or strainer that will remove from the water 98 percent by weight of solids, 74 microns or larger, shall be installed in the intake line between the antisyphon device and the tank. The separator shall be without moving or replaceable parts, shall spin the solid matter through tangential openings in the inner shell, have a pressure drop less than 8 lb/in^2g, and be equipped at its lowest point with a quick-acting ball valve for discharge of the settled matter.

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3.12.4 Method of loading. The method of loading shall allow the pickup hose to encompass an arc of not less than 60° on either side of the vehicle centerline at a horizontal reach not less than 12 feet from vehicle circumference. Debris shall be deposited directly into the vacuum body without going through the compressor blower. When specified (see 6.2), means shall be provided that shall sound an audible alarm when the main body is full.

3.13 Hydraulic boom. Boom shall meet the reach requirements of 3.12.4 and table I shall be provided. When specified (see 6.2), for size 2, an extendable hydraulic boom shall be provided that can be extended to 24 feet from the vehicle centerline and elevate to not less than 25 feet. When retracted the 24 foot boom shall not exceed 14 feet in length. The boom provided shall have a lifting capacity of 1,000 lb at a point 12 feet from the vehicle circumference. Hydraulic powered cylinders and/or electric motors shall be furnished to allow power assisted movement of the boom both vertically and horizontally using a cable mounted remote control station locatable near the pickup hose. Means shall be provided for storage of the control station on the cleaner.

3.14 Pickup hose. The pick up hose shall not be less than 6 inches nor more than 8 inches in diameter made of wire reinforced rubber hose. The length of the hose shall reach from the body of the cleaner to the sewer manhole or catch basin to be cleaned at all operating positions. Hose shall be secured to flanged tubes with hose clamps. Extension tubes shall be provided in sufficient length to vacuum from a depth of 25 feet below ground level. Truck mounted racks shall be provided to hold at least 25 feet of the extension tubes. Extension tubes shall be attached to pickup hose by means of quick coupler type over-center clamps. The hose assembly shall be sufficiently tight to prevent a 2 by 2 inch piece of typing paper from being held by suction at any point. When specified (see 6.2), on type II unit, size 2, additional extension tubes shall be provided in sufficient length to vacuum from a depth of 100 feet below ground level, and shall be provided with a flanged "Y" connection and 25 feet of removable wire reinforced rubber or plastic hose. The "Y" connection inlet sections and hose diameter shall be not less than 4 inches or more than 5 inches.

3.15 Auxiliary cleaning water system. An auxiliary cleaning water system shall be provided consisting of a pump, water supply and a hand gun with hose. The auxiliary system shall deliver water to the areas of the catch basin and to the inside of the vacuum box for cleaning debris and sludge. A minimum of 20 gal/min regulated at 600 lb/in^2g shall be available at the nozzle. The hose for the auxiliary cleaning system shall have a burst pressure of at least 2,400 lb/in^2g and a proof pressure of 1,200 lb/in^2g.

3.16 Auxiliary power. If required for proper operation of the cleaner, the cleaner shall be equipped with liquid-cooled compression ignition, two or four stroke diesel power, having hp, torque, and speed characteristics to satisfactorily perform all functional requirements and not exceed the engine manufacturer's recommended rating for continuous duty. Power takeoff for

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engaging and disengaging equipment shall be provided for the necessary operations of the cleaner. If main engine through PTO drive unit is insufficient to operate both vacuum and high pressure functions of unit, an auxiliary diesel engine of sufficient power may be provided. The power equipment shall comply with the applicable EPA Regulations governing control of Air Pollution for New Motor Vehicles and New Motor Vehicle Engines in effect on the date of manufacturer. In addition, equipment destined for California shall comply with State of California Regulations governing Air Pollution Control in effect on the date of manufacture. If an ether starting system in furnished in lieu of a glow plug, it shall be of the measured shot type, controlled from the driver's compartment, with a reservoir of not less than 12 fluid ounces, and shall be inoperative with the engine warm.

3.16.1 Oil filter. Full flow oil filter shall be furnished.

3.16.2 Governor. Engine governor shall be furnished and set and sealed to limit engine to manufacturer's maximum recommended operating speed.

3.16.3 Fuel systems. Fuel systems shall conform to DoT Federal Motor Vehicle Safety Standards. Fuel tank shall have sufficient capacity for 8 hours of normal operation.

3.16.4 Air cleaner. A dry type engine intake air cleaner shall be furnished, and provided with a restrictor indicator gage.

3.16.5 Electrical system. The electrical system shall be 12V. Alternator shall have a minimum rated capacity of 80A. The battery shall be charged and ready for use.

3.16.6 Accessories and equipment. The engine shall be complete with all accessories furnished as standard equipment by the manufacturer. The following minimum equipment shall be furnished.

a. Key-operated ignition switch.
b. Ammeter or charging indicator.
c. Fuel gage.
d. Oil pressure gage or indicator.
e. Engine temperature gage or indicator.
f. Hour meter of the sealed repeating type and register up to at least 9,999.9 hours.
g. Tachometer.
h. Pressure gage for high-pressure pump.

3.17 Blower compressor. A blower or compressor shall be provided. The blower/compressor shall be able to deliver the quantity of air and the velocity to meet the performance requirements of paragraph 3.10.3.

3.17.1 Blower compressor inlet filter. If a positive displacement blower/compressor is used, an inlet filter shall be provided to prevent ingestion of foreign matter. The size, capacity and filtration performance

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shall meet the blower-compressor manufacturer's particle size filtration requirements. A water mist spray system only is not acceptable. The filtration system shall operate in both wet and dry vacuuming operations.

3.18 High pressure pump. The high pressure pump shall deliver as a minimum the quantity of water as specified in 3.10.1. A system shall be provided to vary the flow from zero to maximum capacity by the operator at the control panel or station. Controls shall be furnished whereby the operator can start and stop the pump. Relief valves shall be furnished which can be adjusted to any desired pressure setting for water delivery. The water side of the pump shall have all surfaces in contact with water manufactured from one of the following materials: hard chrome, stainless steel, polyurethane, teflon, rubber, or high strength gray iron. The pump shall have adjustable packing glands or renewable seals and be protected by a suction line filter.

3.19 Hose assembly. A hose reel assembly shall be furnished consisting of a hose reel, 500 feet of hose, and hydraulic motor with controls. The hose reel shall be power driven both forward and reverse by a hydraulic motor. The controls for the motor shall incorporate a control device to regulate the rotational speed of the reel in both directions. A footage counter shall be provided that is visible and accessible to the operator while operating the machine. All controls for operating the reel assembly shall be accessible to the operator from any operator position. The reel capacity shall be a minimum of 500 feet of 1-inch inside diameter hose. The hose reel spool radius shall not be less than the specified bend radius of the hose furnished and shall allow the first layer to spool firmly. The 1-inch inside diameter hose shall have an operating pressure of 2,000 lb/in^2g. The hose shall conform to SAE 100R8 of SAE J517. A shoe for guiding the hose into the drainage line shall be furnished.

3.20 Tool box. A tool box of sufficient size to stow flanges, collars, quick clamps, gaskets, nozzles, hand pressure gun, portable light, control stations and all other loose hand tools or devices shall be provided. The tool box shall have dividers to separate the various items, shall be weatherproof all steel construction of not less than a nominal 0.0747-inch (U.S. revised standard gage No. 14) thick steel with a hinged lid and a trunk drawbolt to keep the lid secure when vibrated. The tool box shall be mounted in a protected, accessible location.

3.21 Operating and equipment tools. The following operating and equipment tools shall be furnished:

	Item	Quantity
a.	Bench vice with 4-inch jaws.	1
b.	Hose clamping block for 1-inch hose.	1
c.	Round leaf nozzle.	1
d.	If supplied as separate items Quick clamps for suction tube.	10

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Item		Quantity
e.	If supplied as separate items Gaskets-rubber for suction tube.	10
f.	Basin retriever.	1
g.	Nozzle for 1-inch hose with spray angle between 30° and 40° hardened to 50 Rockwell C.	2 pcs
h.	Nozzle for 1-inch hose with spray angle between 10° and 20° hardened to 50 Rockwell C.	2 pcs
i.	Storm sewer nozzle for 1-inch hose equipped with replaceable spray jets for use in 18- through 36-inch pipe.	1
j.	Nozzle pipe assembly.	1
k.	Hand light 12V spot.	1
l.	Lid lifter hook.	1
m.	Offset manhole roller.	1
n.	Pick.	1
o.	Hydrant wrench..	1
p.	Hydraulic (water) gear driven root cutter kit, 6-inch through 12-inch pipe.	1
q.	If supplied as separate item tools to repair high pressure water pump, check valves and packings	1 kit

3.22 Rustproofing. The vehicle shall be rustproofed in accordance with FED-STD-297.

3.23 Lubrication. The lubricating points shall be easily visible and accessible. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high pressure lubricating equipment, 1,000 pound-force per square inch (psi) or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location.

3.24 Instruction plates. The cleaner shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment. Plates shall be securely affixed to the equipment with nonferrous screws or bolts of not less than 1/8-inch diameter.

3.25 Identification plate. The contracting officer will furnish to the Government inspector the required identification plates. The contractor will be required to stamp the necessary data in the blank spaces thereon and securely affix said plates in a conspicuous place on each unit, assembly or subassembly, and parts as directed by the Government inspector. Nonferrous

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screws, rivets, or bolts of not less than 1/8-inch in diameter shall be used to affix the plates. Nomenclature shall be "TRUCK, CLEANER, CATCH BASIN, VACUUM-PRESSURE JET".

3.26 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be No. 13538, yellow, conforming to FED STD 595. Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.27 Servicing and adjusting. Prior to acceptance of the cleaner by the Government, the contractor shall service and adjust the cleaner for immediate operational use as required in the operator's manual. The servicing and adjusting shall include at least the following:

a. Inflation of all tires.
b. Adjustment of brakes when required.
c. Proper functioning of all lighting and electrical systems.
d. Wheel alinement when required.
e. Adjustment of engine to include tune-up when required.
f. Complete lubrication with grades of lubricants recommended for ambient temperature at the delivery point.
g. Cooling system filled to capacity with a clean solution of equal parts by volume of water and antifreeze (ethylene glycol).

The cleaner shall be conspicuously tagged to identify the lubricants and their temperature range.

3.28 Workmanship.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on cleaner when a first article is required (see 3.2 and 6.2). This inspection shall include the examination of 4.4 and the tests of 4.5. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the tests of 4.4, and the packaging inspection of 4.6. This inspection shall be performed on the samples selected in accordance with 4.3.

4.3 Examination. Each cleaner shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 Tests. The cleaner shall be tested in accordance with 4.4.1 through 4.4.8. Nonconformance to the requirements as specified in section 3 shall constitute failure and cause for rejection. If a positive displacement blower/compressor is used, prior to testing, the contractor shall show evidence from the blower/compressor manufacturer specifying the recommended, maximum size particles allowable for ingestion by the blower/compressor during normal service.

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4.4.1 Operational test. The complete cleaner shall be operated a sufficient period of time to demonstrate the design and performance to accomplish the following:

4.4.2 Hydraulic rodder test. The cleaner, with all equipment stowed in the normal transport position(s), and a two person crew, shall move near a covered manhole catch basin not less than 25 feet deep, set up and propel the high pressure hose through an 8 inch diameter unobstructed sewer line for a distance of not less than 400 feet while simultaneously vacuuming all sludge, debris, and cleaning water into the body of the vehicle. The high pressure hose shall self-propel the full 400 feet without manual assistance. After restowing all equipment/accessories to their normal position for travel, the vehicle shall then move at least 400 feet and stop. The total elapsed time for this complete operation shall exceed 17 minutes. Exceeding the 17 minutes elasped time shall be cause for rejection.

4.4.3 Loading wet and dry material test. Using normal operating procedures, the manufacturer shall demonstrate the compressor performance described in 3.10.3 for the type and size of unit specified (see table I) by loading not less than 1 YD^3 each of wet and dry material. Dust emissions for size 2 units shall not exceed requirements of 3.12.1. Failure to load the specified minimum volume of material within time limits as specified in table I shall be cause for rejection. Using a verifible, laboratory approved method, air emissions shall be sampled during each phase including different material and hose combinations. Failure of any part of this test shall be cause for rejection.

4.4.4 Loading light material for size 2 only. Using normal operating procedures, the manufacturer shall demonstrates the blower compressor performance to load light material, leaves, sawdust, etc., to the requirements of 3.10.4. Failure of any cleaner to pass this test shall be cause for rejection.

4.4.5 Loading wet and dry material when "Y" connection is specified for type II only. From along the side of an open pit or from an elevated position and 100 feet below cleaner level effectively vacuum all sludge, debris and water into the body of the truck with the "Y" conntection and the two 25-foot flexible hoses provided under 3.14 attached to the extension piping and located at the pit bottom.

4.4.6 Inlet filter effectiveness test. For cleaners equipped with a positive displacement blower/compressor, filter effectiveness shall be verified. The size of particulate material entrained in the air, downstream of the inlet filter shall be compared to the blower/compressor manufacturer's maximum allowable value. This test may be done concurrently with 4.4.3, if done separately, materials and vacuuming times shall be as defined in 3.10.3. If particles larger than the maximum are detected, this shall be cause for rejection.

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4.4.7 Pump delivery test. The pump shall be tested for compliance with the performance requirements specified in 3.10.1. Failure to pass the requirements shall be cause for rejection of the unit.

4.4.8 Road test. The first article cleaner shall be examined and road tested by the contractor to verify the grade and speed requirements of 3.10.5. The cleaner, loaded with full payload and with all equipment shall be driven at speeds up to at least 50 mph, for a minimum distance of 50 miles. The distance traveled shall consist of highway, rough dirt roads, partly surfaced gravel roads, puddles, and sharp turns. For each 25 miles of travel at least one sudden stop shall be accomplished. Upon completion of the road test, all equipment shall be thoroughly examined. The equipment shall show no evidence of damage; all compartment(s) and cab doors shall properly open and shut and there shall be no evidence of racking of the body.

4.5 Production sample. Upon acceptance of the first article, the first article shall remain at the manufacturing facility as a production sample, and shall be reconditioned, including replacement of abnormally worn parts and paint touch-up or repainting, prior to delivery to enable it to be accepted as a contract item. The contractor shall maintain the first article in a serviceable condition for the duration of the contract.

4.6 Packing inspection. The preservation-packing of the cleaner shall be examined to determine compliance with the requirements of section 5 of this specification.

5. PACKAGING

5.1 Preservation/packing. Unless otherwise specified (see 6.2), the complete cleaner shall be preserved and packed in accordance with the supplier's standard practice. When specified (see 6.2), the complete cleaner shall be preserved and packed in accordance with mobile requirements of MIL-STD-281, with the level of preservation and packing required, (see 6.2).

6. NOTES

6.1 Intended use. Type I cleaner is intended to clean drainage lines to lengths of 400 feet and diameters from 6 inches to 36 inches and remove the sludge and debris by vacuum from the catch-basis to at least 25 feet below ground. Type II cleaner is similar to type I with the added function of removing sludge and debris from dry-dock settling pens and basins to a depth of 100 feet through use of the extended vacuum tube and two 25 foot long wire reinforced rubber hoses.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Size required (see 1.2).
c. When first article is required (see 3.2 and 6.6).

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d. For type II, when inside of body shall be manufactured from a corrosion-resistant steel compatible with a salt water environment equal to ASTM A167, type 316 (see 3.12.1).
e. When alarm is required (see 3.12.4).
f. When 24 foot extenable hydraulic boom is required (see 3.1.13).
g. When additional extension tubes, flanged "Y" connection, and hoses are required (see 3.14).
h. If preservation/packing is other than specified (see 5.1).
i. When preservation/packing in accordance with MIL-STD-281 is required, and the level of preservation/packing required (see 5.1).

6.3 Definitive specification part number. The specification part number is a definitive part number which corresponds to the types and sizes of cleaners covered by this specification and defines the requirements of the options presented under this specification. The specification number, and the type and size code number are combined to form the definitive specification part number.

6.3.1 Cataloging data. For cataloging purposes, part numbers for the cleaner are assigned as follows:

	M29197	11
Specification part number.		
Type and size code number (see 6.3.2).		

6.3.2 Type and size. The type and size of the cleaner (see 1.2) are identified by a two-digit number (see table II).

TABLE II. Code number to type and size.

Type and size	Code number
Type I Size 1	11
Type I Size 2	12
Type II Size 2	22

6.4 Data requirements. When this specification is used in an aquisition and data are required to be delivered, the data requirements shall be developed as specified by an approved Data Item Description (DD Form 1664)

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and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data shall be delivered by the contractor in accordance with the contract or purchase order requirements.

6.5 Subject term (key word) listing.

Catch basin.
Pump.
Sewer.
Truck.
Water tank.

6.6 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

Custodian:
Navy - YD

Preparing activity:
Navy - YD

Project 2320-N365

2320-N365

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-29197A(YD)	MILITARY SPECIFICATION – TRUCK, CLEANER, CATCH BASIN, VACUUM-PRESSURE JET, DED

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITIONS OBSOLETE

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